

Application Protocols

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Application protocols in network stack

Application Layer

- HTTP/HTTPS
 - Websockets
 - MQTT (Message Queuing Telemetry Transport)
 - AMQP (Advanced Message Queuing Protocol)
- These Protocols are used:
- Define the message formats and structures
 - Request/ reply patterns
 - Management of connections
 - Error handling

Transport Layer

TCP

UDP

Network Layer

IP (IP4/IP6)

Datalink Layer

Ethernet

Wifi

Bluetooth

Physical Layer

Hyper Text Transfer Protocol(HTTP)

Both HTTP requests and responses share a similar structural foundation:

a **start-line**, a set of **headers**, an **empty line**, and an optional **body**

GET /index.html HTTP/1.1	<-- Request Line
Host: www.example.com	<-- Header
User-Agent: Mozilla/5.0	<-- Header
Accept-Language: en-US	<-- Header
	<-- Blank Line (Carriage Return and Line Feed (CRLF))
[Optional Request Body Data]	<-- Message Body

HTTP Request Format

An HTTP request is sent by a client (e.g., a browser) to a server to trigger an action.

- **Request Line:** The first line, which includes:
 - **Method:** The action to perform, such as GET (retrieve data) or POST (send data).
 - **Request Target:** Usually a URL or path to the resource (e.g., /index.html).
 - **HTTP Version:** The version being used (e.g., HTTP/1.1).
- **Request Headers:** Key-value pairs providing metadata about the client or the desired resource format (e.g., Host: example.com, User-Agent: Mozilla/5.0).

HTTP Request Format

- **Empty Line:** A mandatory blank line indicating the end of the headers.
- **Request Body:** Optional data sent to the server, typically used in **POST** or **PUT requests** (e.g., form data or JSON).

Example Request:

GET /hello.htm HTTP/1.1

Host: www.example.com

Accept-Language: en-us

[Empty Line]

HTTP Response Format

An HTTP response is the server's answer to the client's request.

- **Status Line:** The first line, containing:
 - **HTTP Version:** The protocol version (e.g., HTTP/1.1).
 - **Status Code:** A 3-digit number indicating the outcome (e.g., 200 for success, 404 for not found).
 - **Reason Phrase:** A brief human-readable description of the status code (e.g., OK).
- **Response Headers:** Metadata about the server and the returned resource (e.g., Content-Type: text/html, Server: Apache).
- **Empty Line:** A mandatory blank line separating headers from the body.

HTTP Response Format

- **Response Body:** The actual content requested, such as an **HTML document, image, or JSON data**.

Example Response:

HTTP/1.1 200 OK

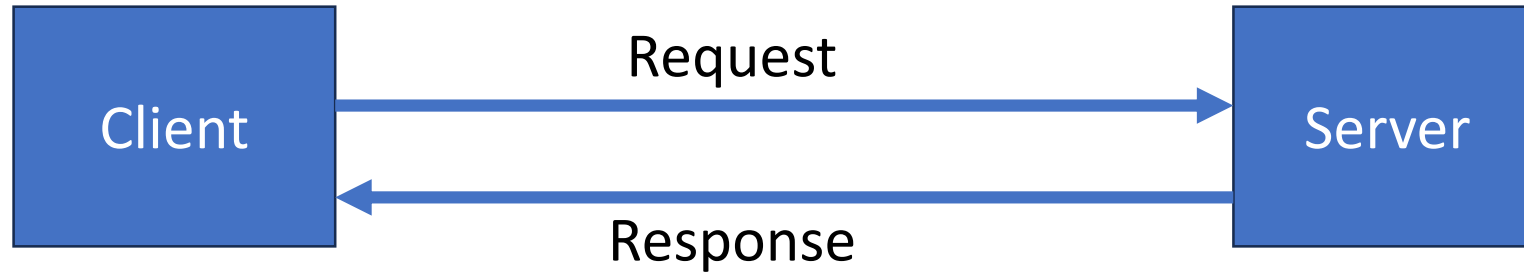
Content-Type: text/html

Content-Length: 128

[Empty Line]

<html>...requested content...</html>

HTTP Protocol



HTTP Methods:

GET: Retrieve Data

POST: Create Data

PUT: Update Data

DELETE: Remove Data

PATCH: Partial Update

Status Codes:

2xx: Success

3xx: Redirection

4xx: Client Error

5xx: Server Error

200: OK

Common Headers:

Content-Type

Authorization

Accept

Cache-Control

User-Agent

HTTPS

- HTTPS= HTTP+TLS/SSL Encryption

Protect data in transit from eavesdropping

- TLS stands for **Transport Layer Security**. It is the most widely used cryptographic **protocol designed to provide secure, encrypted communications** over a computer network, such as the internet.
- The full form of SSL is **Secure Sockets Layer**. It is a standard digital security technology that establishes an encrypted **connection between a user's web browser and a web server**

Benefits

- Data Encryption
- Data Integrity
- Authentication
- Search Engine Optimization(SEO) benefits

Risk without HTTPS

Man-in-the-middle attacks

Data Tampering

Information Theft

Loss of User trust

- Thank You